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Street lamp holder

Abstract

A street lamp holder, including a housing, in which an accommodation cavity, a first sealing groove around the accommodation cavity and a second sealing groove located in the accommodation cavity being formed; a light source accommodated in the accommodation cavity; an optical module accommodated in the accommodation cavity and covering an outer side of the light source; and a sealing member accommodated in the first sealing groove and/or the second sealing groove; the first sealing groove is arranged around the outer side of the light source, and when the sealing member is accommodated in the first sealing groove, a sealing cavity is formed between the accommodation cavity and the housing; the second sealing groove is located between the light source and the first sealing groove, and when the sealing member is accommodated in the second sealing groove, the sealing cavity is formed between the accommodation cavity and the optical module.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application is based upon and claims the priority of PCT patent application No. PCT/CN2022/112934 filed on Aug. 17, 2022 which claims priority to the Chinese patent application No. 202122022814.3 filed on Aug. 26, 2021, the entire contents of which are hereby incorporated by reference herein for all purposes.

TECHNICAL FIELD

(2) The present disclosure relates to a street lamp holder, which belongs to the field of lighting equipment.

BACKGROUND

(3) In the field of lighting, Light-emitting diode (LED) street lamps are classified into types of plastic lenses, glass lenses, glass cover plus reflectors, glass cover plus lenses, etc., and the combination of plastic lenses with glass cover plus lenses is more widely used.

SUMMARY

(4) The present disclosure provides a street lamp holder.

(5) The present disclosure provides a street lamp holder which may include: a housing, where an accommodation cavity, a first sealing groove arranged around the accommodation cavity, and a second sealing groove located in the accommodation cavity may be internally formed in the housing; a light source, where the light source may be accommodated in the accommodation cavity; an optical module, wherein the optical module may be also accommodated in the accommodation cavity and may cover an outer side of the light source; and a sealing member.

(6) The sealing member may be limited and accommodated in the first sealing groove and/or the second sealing groove; the first sealing groove may be arranged around the outer side of the light source, and in a case that the sealing member is limited and accommodated in the first sealing groove, a sealing cavity may be formed between the accommodation cavity and the housing; the second sealing groove may be located between the light source and the first sealing groove, and in a case that the sealing member is limited and accommodated in the second sealing groove, the sealing cavity may be formed between the accommodation cavity and the optical module.

(7) It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is an exploded view of a street lamp holder according to a first example of the present disclosure.

(2) FIG. 2 is a schematic structural diagram of a housing in FIG. 1.

(3) FIG. 3 is a partially enlarged view of the housing in FIG. 2.

(4) FIG. 4 is a schematic structural diagram of a sealing member in FIG. 1.

- (5) FIG. 5 is a cross-sectional view of the street lamp holder in FIG. 1.
- (6) FIG. 6 is a partially enlarged view of a sealed connection in FIG. 5.
- (7) FIG. 7 is an exploded view of a street lamp holder according to a second example of the present disclosure.
- (8) FIG. 8 is an exploded view of an optical module in FIG. 7.
- (9) FIG. 9 is a schematic diagram illustrating various arrangements of lenses in FIG. 8.
- (10) FIG. 10 is a cross-sectional view of the street lamp holder in FIG. 7.
- (11) FIG. 11 is a partially enlarged view of a sealed connection in FIG. 10.
- (12) FIG. 12 is an exploded view of a street lamp holder according to a third example of the present disclosure.

DETAILED DESCRIPTION

- (13) In order to make the purpose, technical solution and advantages of the present disclosure clearer, the present disclosure will be described in details below in conjunction with the accompanying drawings and examples.
- (14) It is to be noted that, in order to avoid obscuring the present disclosure due to unnecessary details, only the structures and/or processing steps closely related to the technical solution of the present disclosure are shown in the drawings, and other details not really matter to the present disclosure are omitted.
- (15) It is to be noted that the terms “comprise/comprising”, “include/including” or any other variations thereof are intended to cover a non-exclusive inclusion such that a process, a method, an article or an apparatus including a set of elements includes not only those elements, but also includes other elements not expressly listed, or also includes elements inherent in such process, method, article or apparatus.
- (16) Some street lamp facilities are mainly based on project bidding, which involves many customized requirements. In order to meet these customized requirements, it is necessary to develop the product functions to satisfy more application requirements.
- (17) As shown in FIG. 1 to FIG. 6, it is a first example of a street lamp holder **100** of the present disclosure, and the street lamp holder **100** is used for lighting outdoors. In this example, the street lamp holder **100** includes a housing **1**, a sealing member **2**, a light source **3**, an optical module **4**, a glass **5** and a connecting member **6**; an accommodation cavity **14**, a first sealing groove **11** arranged around the accommodation cavity **14** and a second sealing groove **12** located in the accommodation cavity **14** are internally formed in the housing **1**; the light source **3** and the optical module **4** are both accommodated in the accommodation cavity **14** of the housing **1**, and the glass **5** is connected to the housing **1** through the connecting member **6**, so that the glass **5** is fixedly mounted with the housing **1**, the optical module **4** covers an outer side of the light source **3**, and the sealing member **2** is arranged in the first sealing groove **11**, thereby realizing a sealed connection between the optical module **4** and the housing **1**, and realizing waterproof and dustproof effect of the street lamp holder **100**.
- (18) As shown in FIG. 2 and FIG. 3, the housing **1** is provided with a heat dissipation assembly **13**, which is used to dissipate heat generated by the light source **3** during a working process, so that the street lamp holder **100** would not be damaged due to excessively higher temperature, thereby prolonging a service life of the street lamp holder **100**. In this example, the accommodation cavity **14**, the first sealing groove **11** and the second sealing groove **12** are all formed by inwardly depressing from an outer surface of the housing **1** at a side away from the heat dissipation assembly **13**; the light source **3** is accommodated in the accommodation cavity **14** and is fixedly mounted with the housing **1** at a bottom of the accommodation cavity **14**, and the sealing member **2** is limited and accommodated in the first sealing groove **11**.
- (19) Specifically, the first sealing groove **11** surrounds an outer side of the light source **3**, the second sealing groove **12** is located between the light source **3** and the first sealing groove **11**, and the second sealing groove **12** is formed by at least two annular grooves **121** communicated with

each other; the light source **3** is arranged in and surrounded by each of the at least two annular grooves **121**. That is to say, the annular groove **121** passes through the middle of the light source **3** to divide the light source **3** and surround the light source **3**, so that the light source **3** can be sealed during assembling. In the present disclosure, by providing the first sealing groove **11** and the second sealing groove **12** on the housing **1**, the multi-functionalization of the housing **1** can be realized to meet the needs of multiple customizations.

(20) The light source **3** is composed of at least two LED lamp boards **31** to be matched with the number of the annular grooves **121**, so that a periphery of each LED lamp board **31** is wrapped by the annular groove **121** to ensure the sealing effect. The bottom of the housing **1** is located between two adjacent LED lamp boards **31**, and is provided with a rib **122**, which separates two adjacent LED lamp boards **31** and also separates two adjacent annular grooves **121**. In addition, notches **123** are also provided at both ends of the rib **122**, and the notches **123** are used to connect two adjacent annular grooves **121**, so as to realize the communication among a plurality of annular grooves **121** and finally form a complete second sealing groove **12**.

(21) Further, the light source **3** in the present disclosure is composed of three LED lamp boards **31**, the three LED lamp boards **31** are evenly distributed in the accommodation cavity **14** and have the same size and the same shape, which is a rectangular shape. Of course, in some other examples, the number, size and shape of the LED lamp boards **31** can be adjusted according to actual conditions, which is not limited here.

(22) The housing **1** is also provided with four fixing holes **15** and several mounting holes **16**; the four fixing holes **15** are distributed in pairs on two long sides of the housing **1**, and the mounting holes **16** are provided on ribs **122** and are evenly distributed along the direction of the short sides of the housing **1**. Preferably, each rib **122** is provided with two mounting holes **16**, and each notch **123** is also correspondingly provided with a mounting hole **16**; of course, in some other examples, both the number and positions of the fixing holes **15** and the mounting holes **16** as arrangement can be set according to actual situations, which is not limited here.

(23) In this example, the fixing holes **15** are arranged between the second sealing groove **12** and the light source **3**; specifically, the fixing holes **15** are arranged on the LED lamp boards **31** located at both ends of the light source **3**, and are located at the middle positions of the short sides of the LED lamp boards **31**. Of course, in some other examples, both the number and positions of the fixing holes **15** can be set according to actual situations, which is not limited here.

(24) As shown in FIG. 4 and FIG. 6, the sealing member **2** is arranged in an annular shape and is matched with the first sealing groove **11** in both shape and size. Of course, a top portion of the sealing member **2** can also be protruded beyond the first sealing groove **11**. When the sealing member **2** is limited and accommodated in the first sealing groove **11**, a sealing cavity **141** can be formed between the accommodation cavity **14** and the housing **1** to completely seal the street lamp holder **100** and realize waterproof and dustproof effect of the street lamp holder **100** as a whole. The sealing member **2** is provided with bumps **21**, which extend from the sealing member **2** towards the light source **3** and are arranged in a circular shape; the number and the positions of the bumps **21** correspond to the number and the positions of the fixing holes **15**. The bump **21** is provided with a through hole **22**, so that the connecting member **6** can be used to pass through the through hole **22** to be connected with the fixing hole **15** to achieve a fixed connection between the sealing member **2** and the housing **1**.

(25) The optical module **4** includes lenses, and the lenses **4** are arranged to cover an outer side of the light source **3** so that light emitted by the light source **3** can be emitted to the outside. For the convenience of description, each lens **4** is divided into two parts, i.e., lens pieces **41** and a base board **42**, which will be explained in details below. The lens pieces **41** are integrally formed with the base board **42**; the lens pieces **41** are formed on the base board **42** and are protruded outwards from the base board **42**. Generally, the lens piece **41** is transparent, and the shape of the lens piece **41** can be one or more of a circular shape, an elliptical shape, a dumbbell-like shape and a crescent

shape. The lens pieces **41** can be arranged on the base board **42** in an array, in a staggered manner or in a centrally symmetrical manner. Of course, in some other examples, the shape and the arrangement of the lens pieces **41** can be adjusted according to actual conditions, which is not limited here.

(26) In this example, at least two optical modules **4** are provided, and the at least two optical modules **4** correspond to the at least two annular grooves **121**; of course, the shapes and arrangements of the lens pieces **41** on the at least two optical modules **4** can be the same or different, depending on actual situations.

(27) The glass **5** is arranged in a rectangular shape to be matched with the shape of the first sealing groove **11**, but the size of the glass **5** is larger than the size of the first sealing groove **11**, so that the glass **5** can cover the first sealing groove **11** completely; in this case, the sealing cavity **141** is formed between the accommodation cavity **14** and the glass **5**. The glass **5** is provided with connecting holes **51** corresponding to the fixing holes **15**, so that when the glass **5** is assembled, the connecting members **6** can be used to pass through the connecting holes **51** to be locked and fixed with the fixing holes **15**, thereby achieving a fixed connection between the glass **5** and the housing **1**. Preferably, the connecting members **6** are screws, and the fixing holes **15** are screw holes, but the present disclosure should not be limited thereto.

(28) When assembling, firstly, the sealing member **2** is placed into the first sealing groove **11**, with the through holes **22** on the sealing member **2** being corresponding to the fixing holes **15** at the bottom of the housing **1**, and then the glass **5** is pressed onto the housing **1**, so that the sealing member **2** is squeezed and deformed; finally, the connecting members **6** are caused to pass through the connecting holes **51** on the glass **5** and the through holes **22** on the sealing member **2** in sequence, until the connecting members **6** are locked and fixed with the fixing holes **15** at the bottom of the housing **1**; in this way, the sealed connection between the glass **5** and the housing **1** can be realized, and the light source **3** is arranged in the sealing cavity **141** formed and surrounded by the glass **5** and the housing **1**, thereby realizing the waterproof and dustproof effect of the street lamp holder **100**.

(29) As shown in FIGS. **7** to **11**, it is the second example of the street lamp holder **100'** of the present disclosure. The structure of the street lamp holder **100'** in the second example is substantially the same as the structure of the street lamp holder **100** in the first example, with the difference merely in that the glass is omitted, and the structure of the optical module is modified to be matched with the second sealing groove **12**. The following part of the description will not describe the same structure any more, but mainly describe the differences in details.

(30) In the second example, the optical module **4'** includes a substrate **43** and lenses **4** located on the substrate **43**, and the lenses **4** cover the outer side of the light source **3**. The shape and arrangement of the lenses **4** are the same as those in the first example, which is not repeated herein. The substrate **43** is provided with perforations **431** corresponding to the mounting holes **16**, so that the connecting members **6** can be used to pass through the perforations **431** to be locked and fixed with the mounting holes **16**, thereby achieving a locked connection between the optical module **4'** and the housing **1**. Preferably, at least two optical modules **4'** are provided and correspond to at least two annular grooves **121**, side edges of substrates **43** of two adjacent optical modules **4'** overlap with each other, and the perforations **431** on the two overlapped substrates **43** correspond to each other, so that the connecting members **6** can be used to pass through the perforations **431** on both of the two substrates **43** at the same time to be connected with the mounting holes **16**, thereby realizing a fixed connection between the two adjacent optical modules **4'**.

(31) The sealant **2'** is liquid sealant, the sealant **2'** can fully fill the second sealing groove **12** and can connect the optical module **4'** with the housing **1** in a sealed manner after curing, and a sealing cavity **141'** is formed between the accommodation cavity **14** and the optical module **4'**, so that the light source **3** is sealed in the sealing cavity **141'**, and the waterproof and dustproof effect of the light source **3** is achieved. Of course, in some other examples, the sealing member **2'** can also be a

solid sealing ring; in this case, the size of the sealing ring needs to be matched with the size of the second sealing groove **12**, so that the sealing ring can be clamped in the second sealing groove **12** to achieve the waterproof and dustproof effect of the street lamp holder **100'**.

(32) When assembling, the sealant **2'** is injected into the second sealing groove **12** by an automatic gluing device, then the optical module **4'** is pressed onto the housing **1**, and the connecting members **6** are caused to pass through the perforations **431** to be fixedly locked with the mounting holes **16**, so that a sealed connection between the optical module **4'** and the housing **1** can be realized after the sealant **2'** is cured in the second sealing groove **12**, thereby further achieving the waterproof and dustproof effect of the street lamp holder **100'**.

(33) As shown in FIG. **12**, it is the third example of the street lamp holder **100''** of the present disclosure, which is a combination of the first example as shown in FIG. **1** with the second example as shown in FIG. **7**; that is, in the third example, the street lamp holder **100''** includes the housing **1**, the light source **3**, the glass **5**, the sealing member **2**, the sealing member **2'**, and the optical module **4'**, with the purpose of further increasing the sealing effect of the street lamp holder **100''**.

(34) When assembling, the sealant **2'** is injected into the second sealing groove **12** by an automatic gluing device, then the optical module **4'** is pressed onto the housing **1**, and the connecting members **6** are caused to pass through the perforations **431** to be fixedly locked with the mounting holes **16**, so that a sealed connection between the optical module **4'** and the housing **1** can be achieved after the sealant **2'** is cured in the second sealing groove **12**. Subsequently, the sealing member **2** is placed in the first sealing groove **11**, with the through holes **22** being corresponding to the fixing holes **15**; then the glass **5** is pressed onto the housing **1**, with the connecting holes **51** being corresponding to the through holes **22** and the fixing holes **15**; and then the connecting members **6** are caused to pass through the connecting holes **51** and the through holes **22** sequentially to be fixedly locked and connected with the fixing holes **15**; in this case, under a mutual compression between the glass **5** and the housing **1**, the sealing member **2** is squeezed and deformed to realize a sealed connection between the glass **5** and the housing **1**. That is to say, the sealant **2'** and the second sealing groove **12** form a first layer of sealing structure, and the sealing member **2** and the first sealing groove **11** form a second layer of sealing structure. Such a double-layered sealing structure further improves the waterproof and dustproof effect of the street lamp holder **100''**.

(35) To sum up, in the street lamp holders **100**, **100'**, **100''** of the present disclosure, the first sealing groove **11** and the second sealing groove **12** are provided on the housing **1**, so that a corresponding sealing groove and a corresponding optical module can be selected depending on different application requirements, which not only can reduce development costs of products but also can satisfy application requirements under multiple situations. At the same time, the lenses **4** are designed to have various shapes and different arrangements, which further increases the product capacity of the street lamp holders **100**, **100'**, **100''**.

(36) The objective of the present disclosure is to provide a street lamp holder, which not only has low development cost, but also has strong expansibility.

(37) In order to achieve the above objective, the present disclosure provides a street lamp holder, including: a housing, wherein an accommodation cavity, a first sealing groove arranged around the accommodation cavity, and a second sealing groove located in the accommodation cavity are internally formed in the housing; a light source, wherein the light source is accommodated in the accommodation cavity; an optical module, wherein the optical module is also accommodated in the accommodation cavity and covers an outer side of the light source; and a sealing member, wherein the sealing member is limited and accommodated in the first sealing groove and/or the second sealing groove; the first sealing groove is arranged around the outer side of the light source, and in a case that the sealing member is limited and accommodated in the first sealing groove, a sealing cavity is formed between the accommodation cavity and the housing; the second sealing groove is located between the light source and the first sealing groove, and in a case that the sealing member

is limited and accommodated in the second sealing groove, the sealing cavity is formed between the accommodation cavity and the optical module.

(38) As a further improvement of the present disclosure, the optical module includes a lens, the street lamp holder further includes a glass and a connecting member, the lens covers the outer side of the light source, the sealing member is accommodated in the first sealing groove, and the glass is in contact with the housing through the sealing member and is locked and fixed with the housing through the connecting member so that the sealing cavity is formed between the accommodation cavity and the glass.

(39) As a further improvement of the present disclosure, the housing is provided with a fixing hole, and both the sealing member and the glass are provided with a through hole corresponding to the fixing hole, and the connecting member passes through the through hole to be locked and fixed with the fixing hole.

(40) As a further improvement of the present disclosure, the sealing member is provided with a bump, and the through hole is arranged on the bump.

(41) As a further improvement of the present disclosure, the optical module includes a substrate and a lens located on the substrate, the street lamp holder further includes a connecting member, the lens covers the outer side of the light source, the sealing member is accommodated in the second sealing groove, and the optical module is in contact with the housing through the sealing member and is locked and fixed with the housing through the connecting member.

(42) As a further improvement of the present disclosure, the housing is provided with a mounting hole, the substrate is provided with a perforation corresponding to the mounting hole, and the connecting member passes through the perforation to be locked and fixed with the mounting hole.

(43) As a further improvement of the present disclosure, the sealing member is a sealant, the second sealing groove is formed by at least two annular grooves communicated with each other, the light source is provided in and surrounded by each of the at least two annular grooves, and the sealant fully fills the second sealing groove to connect the optical module and the housing in a sealed manner upon the sealant being cured.

(44) As a further improvement of the present disclosure, at least two optical modules are provided and correspond to the at least two annular grooves, side edges of substrates of two adjacent optical modules overlap with each other so that the connecting member passes through the perforations on both of two substrates to fixedly connect the two adjacent optical modules.

(45) As a further improvement of the present disclosure, the street lamp holder further includes a glass and a connecting member, wherein the sealing member is provided in each of the first sealing groove and the second sealing groove, and the optical module is in contact with the housing through the sealing member in the second sealing groove and is fixedly connected to the housing through the connecting member; the glass is in contact with the housing through the sealing member in the first sealing groove and is fixedly connected to the housing through the connecting member.

(46) As a further improvement of the present disclosure, the optical module includes lenses, and a shape of the lens includes one or more of a circular shape, an elliptical shape, a dumbbell-like shape and a crescent shape; and the lenses are arranged in an array, arranged in a staggered manner or arranged in a centrally symmetrical manner.

(47) The beneficial effects of the present disclosure are as follows: the street lamp holder of the present disclosure is provided with the first sealing groove and the second sealing groove on the housing, so that a corresponding sealing groove and a corresponding optical module can be selected depending on different application requirements, which not only can reduce development costs of products but also can satisfy application requirements under multiple situations.

(48) The present disclosure may include dedicated hardware implementations such as application specific integrated circuits, programmable logic arrays and other hardware devices. The hardware implementations can be constructed to implement one or more of the methods described herein.

Examples that may include the apparatus and systems of various implementations can broadly include a variety of electronic and computing systems. One or more examples described herein may implement functions using two or more specific interconnected hardware modules or devices with related control and data signals that can be communicated between and through the modules, or as portions of an application-specific integrated circuit. Accordingly, the system disclosed may encompass software, firmware, and hardware implementations. The terms “module,” “sub-module,” “circuit,” “sub-circuit,” “circuitry,” “sub-circuitry,” “unit,” or “sub-unit” may include memory (shared, dedicated, or group) that stores code or instructions that can be executed by one or more processors. The module refers herein may include one or more circuit with or without stored code or instructions. The module or circuit may include one or more components that are connected.

(49) The above examples are only used to illustrate the technical solutions of the present disclosure instead of constituting any limitation thereto. Although the present disclosure has been described in details with reference to the examples, one of ordinary skilled in the art should understand that the technical solutions of the present disclosure can be modified or equivalently replaced without departing from the spirit and scope of the technical solutions of the present disclosure.

Claims

1. A street lamp holder, comprising: a housing, wherein an accommodation cavity, a first sealing groove arranged around the accommodation cavity, and a second sealing groove located in the accommodation cavity are internally formed in the housing; a light source, wherein the light source is accommodated in the accommodation cavity; an optical module, wherein the optical module is also accommodated in the accommodation cavity and covers an outer side of the light source; and a sealing member, wherein the sealing member is limited and accommodated in the first sealing groove and the second sealing groove; the first sealing groove is arranged around the outer side of the light source, and in a case that the sealing member is limited and accommodated in the first sealing groove, a sealing cavity is formed between the accommodation cavity and the housing; the second sealing groove is located between the light source and the first sealing groove, and in a case that the sealing member is limited and accommodated in the second sealing groove, the sealing cavity is formed between the accommodation cavity and the optical module.
2. The street lamp holder according to claim 1, wherein: the optical module comprises a lens, and the street lamp holder further comprises a glass and a connecting member, and the lens covers the outer side of the light source, the sealing member is accommodated in the first sealing groove, and the glass is in contact with the housing through the sealing member and is locked and fixed with the housing through the connecting member so that the sealing cavity is formed between the accommodation cavity and the glass.
3. The street lamp holder according to claim 2, wherein the housing is provided with a fixing hole, and both the sealing member and the glass are provided with a through hole corresponding to the fixing hole, and the connecting member passes through the through hole to be locked and fixed with the fixing hole.
4. The street lamp holder according to claim 3, wherein the sealing member is provided with a bump, and the through hole is arranged on the bump.
5. The street lamp holder according to claim 1, wherein the optical module comprises a substrate and a lens located on the substrate, and the street lamp holder further comprises a connecting member, the lens covers the outer side of the light source, the sealing member is accommodated in the second sealing groove, and the optical module is in contact with the housing through the sealing member and is locked and fixed with the housing through the connecting member.
6. The street lamp holder according to claim 5, wherein the housing is provided with a mounting hole, the substrate is provided with a perforation corresponding to the mounting hole, and the

- connecting member passes through the perforation to be locked and fixed with the mounting hole.
7. The street lamp holder according to claim 6, wherein the sealing member is a sealant, the second sealing groove is formed by at least two annular grooves communicated with each other, the light source is provided in and surrounded by each of the at least two annular grooves, and the sealant fully fills the second sealing groove to connect the optical module and the housing in a sealed manner upon the sealant being cured.
8. The street lamp holder according to claim 7, wherein at least two optical modules are provided and correspond to the at least two annular grooves, side edges of two substrates of two adjacent optical modules overlap with each other so that the connecting member passes through the perforations on both of the two substrates to fixedly connect the two adjacent optical modules.
9. The street lamp holder according to claim 1, further comprising: a glass and a connecting member, wherein the sealing member is provided in each of the first sealing groove and the second sealing groove, and the optical module is in contact with the housing through the sealing member in the second sealing groove and is fixedly connected to the housing through the connecting member; the glass is in contact with the housing through the sealing member in the first sealing groove and is fixedly connected to the housing through the connecting member.
10. The street lamp holder according to claim 1, wherein: the optical module comprises lenses, and a shape of the lens comprises one or more of a circular shape, an elliptical shape, a dumbbell-like shape and a crescent shape, and the lenses are arranged in an array, arranged in a staggered manner or arranged in a centrally symmetrical manner.
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